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1 Lecture 1

1.1 Course Overview

This course is titled "Matrix Concentration and Applications." Matrix Concentration is the development of concentration inequalities involving spectral properties of a matrix of random variables. Here are the applications:

- Algorithms: Randomized numerical linear algebra, analysis of algorithms on randomized inputs
- Statistical Estimation: How many samples from a distribution do you need to figure out its parameters?
- Coding theory/combinatorics: Solve extremal girth problems, local codes, Euclidean sections.
- Complexity Theory: Expander graphs (random graphs are Ramanujan...)
- Quantum Information: Quantum query algorithms can be seen through the lens of Matrix concentration.
- Discrepancy theory: How can you take a set of points and split them such that the two points are indistinguishable by some test.

1.2 Scalar Concentration Inequalities

Theorem 1.1 (Khintchine)
Suppose $a_1, \ldots, a_n \in \mathbb{R}$ and $g_1, \ldots, g_n \sim N(0,1)$, iid. Then

$$\mathbb{E}\left[|\sum_i g_i a_i|\right] = \sqrt{\frac{2}{\pi}}\sqrt{\sum_i a_i^2}$$

Suppose $b_1, \ldots, b_n$ are independent uniform Rademacher random variables. Then

$$\frac{1}{\sqrt{2}}\sqrt{\sum_i a_i^2} \leq \mathbb{E}\left[|sum_i b_i a_i|\right] \geq \sqrt{\sum_i a_i^2}$$

Theorem 1.2 (Bernstein)
Let $X_1, \ldots, X_n$ be independent, zero-mean and for all i, $X_i \leq L$ almost surely. Define $\sigma = \sqrt{\sum_{i=1}^n \mathbb{E}\left[X_i^2\right]}$ to be the variance of the sum. Then,

$$\Pr \sum_{i=1} X_i \geq 2t\sigma \leq e^{-t^2}$$

for all $0 \leq t \leq \sigma/2L$.

Theorem 1.3 (Chernoff)
Let $X_1, \ldots, X_n$ be independent Bernoullis, where $\mathbb{E}\left[X_i\right] = p_i$. Define $\mu = \sum_i p_i$. Then, $\forall \delta > 0$,

$$\Pr \sum_i X_i \geq (1+\delta)\mu \leq e^{\frac{-\mu\delta^2}{2+\delta}}$$

and for $0 < \delta < 1$

$$\Pr \sum_i X_i \leq (1-\delta)\mu \leq e^{\frac{-\mu\delta^2}{2}}$$

It turns out all of them reduce to Khintchine...

1.3 Matrix Concentration Inequalities

Theorem 1.4 (Non-Commutative Khintchine)
Let $A_1, \dots, A_n$ be $m \times m$ symmetric matrices. and $b_1, \dots, b_n$ be independent uniform Rademachers or iid $N(0,1)$. Then

$$\Pr\left\|\sum_i b_i A_i\right\|_2 \geq t\sigma \leq 2me^{-t^2/2}$$

where $\sigma^2 = \left\|\sum_i A_i^2\right\|_2$.

Note the similarity to the scalar Bernstein inequality, except we get an extra dependence m on the dimension of the matrix. I.e., with high probability we have that $\|\sum_i b_i A_i\|_2 \sim \sqrt{\log m}\sigma$. One may ask whether this inequality is really tight. Define $A_i = e_i e_i^T$. Then, $\sigma^2 = \left\|\sum_i A_i^2\right\|_2 = \|I\|_2 = 1$. However, $\|\sum_i g_i A_i\|_2 = \max\{g_1, \dots, g_n\}$. Its expectation is $\sim \sqrt{\log m}$.

Consider instead $A_i = diag(-1, +1, \dots)$, where we assign arbitrary signs to each i. Then, $\sum_i b_i A_i = diag(\langle b, v_1\rangle, \dots, \langle b, v_m\rangle)$. Then, each diagonal entry is the sum of random signs. Then, the max of this concentrates to $\sqrt{\log m}$.

If the A_i's commute with each other, then we claim that the spectral norm is fixed to this.

1.4 Graph Sparsification

Let $G = (V, E)$ be a graph. The goal of graph sparsification is to get a new (weighted) graph $G' = (V, E')$, $w : E' \to \mathbb{R}_+$ where $|E'| \ll |E|$ while G' approximates G. Define a cut value as $\mathrm{Cut}_H(S) = \sum_{e \in E(S,\overline{S})} w_H(e)$.

Definition 1.5
(ϵ-sparsifier) $G' \approx_\epsilon G$ if for all sets S, $\mathrm{Cut}_{G'}(S) \in (1+\epsilon)\mathrm{Cut}_G(S)$

Define $L_G = D - A = \sum_{e \in E} w_e L_e$ where $L_{(i,j)} = e_i e_i^T + e_j e_j^T - e_i e_j^T - e_j e_i^T$. Recall $x^T L_G x = \sum_{(i,j) \in E} (x_i - x_j)^2 w_{(i,j)}$, so it is exactly $1/4$ the cut value for $x \in \{+1, -1\}^n$.

Theorem 1.6
There is such a G' with $|E'| = O\left(\frac{n \log n}{\epsilon^2}\right)$.

The main idea is using randomly sampling each edge p_e and weight it $1/p_e$ if you keep it. The first thing you can try is $p_e = p$ for all edges independently. If the min-cut in G is of size k, then it turns out that $p \sim \frac{\log(n)}{\epsilon^2 k}$. will work. Otherwise, $K_{n/2} - - - K_{n/2}$ requires the edge in between to be present. So uniform sampling is not good enough. We will construct spectral sparsifiers, a more strict version of edge sparisification, where we want quadratic forms to be prserved for all x, instead of just boolean.

The idea is to reframe this as a partitioning problem. Start from $E_0 = E$ and pick $E_i \subseteq E_{i-1}$ and $|E_i| \leq \frac{1}{2}|E_{i-1}|$ for all $i \geq 1$.

Definition 1.7
We define normalized edge Laplacian

$$\tilde{L}_e = L_G^{\dagger/2} L_e L_G^{\dagger/2}$$

where $L_G^{\dagger/2}$ is the PSD square root of the pseudo-inverse.

Note that $\sum_e \tilde{L}_e = L_G^{\dagger/2} L_G L_G^{\dagger/2} = I_\top$. We want that $\left|x^T \sum_{e \in E_1} \frac{1}{p_e} L_e x - x^T L_G x\right| \leq \epsilon x^T L_G x$ for all x. This happens if

and only if $\left|x^T \sum_{e\in E_1} \frac{1}{p_e}\tilde{L}_e x - x^T I_\top x\right| \le \epsilon x^T I_\top x$ for all x.

The idealized algorithm is as follows.

Algorithm 1.8
$E_0 = E$, $w^0 = 1$, for $t = 1, \dots, t_0$.

1. Find $s \in \{\pm 1\}^{E_{t-1}}$ minimizing
$$\left\|\sum_{e\in E_{t-1}} s_e w_e^{t-1}\tilde{L}_e\right\|_2$$
2. Set $E_t \subseteq E_{t-1}$ to be all $e \in E_{t-1}$ such that $s_e = +1$.
3. Set $w^t = 2w_e^{t-1}$ if the edge was chosen.

To start analysis, note that

$$\begin{aligned}\left\|\sum_{e\in E_{t_0}} w_e^{t_0}\tilde{L}_e - I_\top\right\|_2 &\le \sum_{i=0}^{t_0-1}\left\|\sum_{E_i} w_e^i \tilde{L}_e - 2\sum_{e\in E_{i+1}} w_e^i \tilde{L}_e\right\|_2 \\ &= \sum_{i=0}^{t_0-1}\left\|\sum_{e\in E_i} s_e^i w_e^i \tilde{L}_e\right\|_2\end{aligned}$$

Lemma 1.9
Suppose that for all e, $\left\|\tilde{L}_e\right\|_2 \le \rho$. Then, $\mathbb{E}\left[\left\|\sum_e s_e\tilde{L}_e\right\|_2\right] \lesssim \rho^{1/2}\sqrt{\log n}$.

Proof 1.10
We use the expectation version of Khintchine.

$$\mathbb{E}\left[\left\|\sum_e s_e\tilde{L}_e\right\|_2\right] \lesssim \left\|\sum_e \tilde{L}_2^2\right\|_2^{1/2}\sqrt{\log n}$$

Furthermore,

$$\left\|\sum_e \tilde{L}_e^2\right\|_2 \le \left\|\sum_e \rho I\tilde{L}_e\right\| \le \rho\left\|\sum_e \tilde{L}_e\right\| = \rho$$

What's the best bound we can get on the $\tilde{L}_e$? $\mathrm{Tr}(\sum_e \tilde{e}L_e) = \mathrm{Tr}(I_\top) = n-1$. Since each $\tilde{L}_e$ is rank one, $\left\|\tilde{L}_e\right\|_2 = \mathrm{Tr}(\tilde{L}_e)$. Thus the sum of their spectral norms is $n-1$. The best bound we can hope for is $\left\|\tilde{L}_2\right\|_2 = \frac{n-1}{m}$. Thus, $\rho \ge \frac{n-1}{m}$. Let's say for the time-being that we had this best case, i.e. $\mathrm{Tr}\left(\sum_e w_e^i\tilde{L}_e\right) \sim n-1$. Then,

$$\left\|w_e^i\tilde{L}_e\right\|_2 \lesssim \sqrt{\frac{n-1}{m_i}}$$

So the total error is roughly $\sqrt{\log n}\left(\sqrt{\frac{n}{m}} + \sqrt{\frac{n}{m_2}} + \dots\right)$. These are geometrically following, so if $m_{t_0} \sim n\log(n)/\epsilon^2$, then it's roughly $\sqrt{n\log n/m_0}$.

We can fix these idealizations as follows. We know that $\sum_e w_e^i\tilde{L}_e \preceq (1+\text{ error})I_\top$, so then the trace is at most $n+$ error, which will be $o(1)$. Call $\rho_e^i = \left\|w_e^i\tilde{L}_e\right\|_2$. Then $\sum_e \rho_e^i \approx n$. By Markov, at least $|E_i|/2$ edges satisfy $\sqrt{\rho_e^i} \le 2\frac{n}{m_i}$. Now we will only sign quantities such that the above holds.

2 Lecture 2

I was out.

3 Lecture 3

3.1 Concentration of Variance

We will show a subgaussian decay tail bound, by showing we can convert from the expectation versions of the bounds to high probability ones.

Note the standard fact $\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var} X_i$ for independent X_i's. Also $\mathrm{Var}(x) = \frac{1}{2}\mathbb{E}_{x,y}(x-y)^2$ for (x, y) as two independent copies of y.

Theorem 3.1 (Efron-Stein)
Suppose $X_1, \ldots, X_n$ are independent, real-valued random variables. Then for all $f : \mathbb{R}^n \to \mathbb{R}$,

$$\mathrm{Var}(f(X_1, \ldots, X_n)) \leq \sum_{i=1}^{n} \mathbb{E}_{x\sim i}\mathrm{Var}_{x_i} f(x)$$

Where we mean

$$\begin{aligned}\mathbb{E}_{x\sim i}\mathrm{Var}_{x_i} f(x) &:= \mathbb{E}_{x_1,\ldots,x_{i-1},x_{i+1},\ldots,x_n}\mathbb{E}_{x_i}(f(x_1, \ldots, x_n) - \mathbb{E}_{x_i} f(x_1, \ldots, x_n))^2 \\ &= \frac{1}{2}\mathbb{E}_{x,y}(f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, y_i, x_{i+1}, \ldots, x_n))^2\end{aligned}$$

Proof 3.2
Define $(\mathbb{E}_i f)(x_1, \ldots, x_{i-1}, \cdot, x_{i+1}, \ldots, x_n) := \mathbb{E}_{x_i} f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n)$. Define $\mathbb{E}_S f = \mathbb{E}_{i_1}\mathbb{E}_{i_2}\ldots\mathbb{E}_{i_{|S|}} f$. Notice that $f = (f - \mathbb{E}_{[1]}f) + (\mathbb{E}_{[1]}f - \mathbb{E}_{[2]}f) + \cdots + (\mathbb{E}_{[n-1]}f - \mathbb{E}_{[n]}f) + \mathbb{E}_{[n]}f$. Notice that $\mathbb{E}_{[n]}f = \mathbb{E}f$. Then, we claim that in the quantity $\mathbb{E}(f - \mathbb{E}f)^2$, the cross terms cancel. For example,

$$\mathbb{E}[(f - \mathbb{E}_{[1]}f)(\mathbb{E}_{[1]}f - \mathbb{E}_{[2]}f)] = \mathbb{E}f\mathbb{E}_{[1]}f - \mathbb{E}f\mathbb{E}_{[2]}f - \mathbb{E}(\mathbb{E}_{[1]}f)^2 + \mathbb{E}(\mathbb{E}_{[1]}f)(\mathbb{E}_{[2]}f) = 0$$

where the first two terms cancel by the tower rule, and same with the second two. Now, $\mathbb{E}(f - \mathbb{E}f)^2 = \sum_{i=0}^{n-1}\mathbb{E}(\mathbb{E}_{[i]}f - \mathbb{E}_{[i+1]}f)^2$. Now, we can apply Jensen's inequality term by term, to get an expecation over the correct variables.

Definition 3.3
Consider $\Sigma \subseteq \Sigma$ and $f : \Sigma^n \to \mathbb{R}$.

$$\begin{aligned}D_i f(x) &= \sup_{y\in\Sigma} f(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n) - \inf_{y\in\Sigma} f(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n) \\ \Delta_i f &:= \sup_{x\in\Sigma^n}(D_i f(x))\end{aligned}$$

Note that if $x \in [a, b]$ almost surely, then, $\mathrm{Var}(x) \leq \frac{1}{4}(b-a)^2$.

Corollary 3.4

$$\mathrm{Var}(f(x_1, \ldots, x_n)) \leq \frac{1}{4} \sum_{i=1}^{n} \mathbb{E}(D_i f(x))^2 \leq \frac{1}{4} \sum_i (\Delta_i f)^2$$

Theorem 3.5
Let $W \in \mathbb{R}^{n \times n}$ be symmetric, independent entries such that $|W_{ij}| \leq K$. Then, $\mathrm{Var}(\lambda_1(W)) \leq 16K^2$.

Corollary 3.6
Any Erdos-Renyi random graph's largest eigenvalue has variance at most 16.

Definition 3.7
$D_i^+ f(x) = \sup_{y \in \Sigma} f(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n) - f(x)$.

Then the corollary before turns into.

$$\mathrm{Var}(f(x_1, \ldots, x_n)) \leq \frac{1}{4} \sum_{i=1}^{n} \mathbb{E}(D_i^+ f(x))^2 \leq \frac{1}{4} \sum_i (\Delta_i^+ f)^2$$

Now we can prove the variance theorem.

Proof 3.8
We now will control $D_{ij}^+ \lambda_1(W)$. Suppose that $W \neq W'$ differ in only the (i, j)th entry. Let v be the top eigenvector of W.

$$\lambda_1(W) - \lambda_1(W') = v^T W v - \max_{\|u\|=1} u^T W' u = v^T W v - v^T W' v \leq 2|(W_{ij} - W'_{ij})||v_i v_j| \leq 4K|v_i v_j|$$

Then by the new corollary we have $\mathrm{Var}(f) \leq 16K^2 \sum_{i,j} v_i^2 v_j^2 = 16K^2$.

Theorem 3.9 (Gaussian Poincare Inequality)
Let $f : \mathbb{R}^n \to \mathbb{R}$ be $\mathcal{C}^1$ and

$$\mathrm{Var}_{x \sim \mathbb{N}(0,I)} f(x) \leq \mathbb{E}\|(\nabla f)(x)\|_2^2$$

Proof. By Efron-Stein we have that

$$\begin{aligned}
\mathrm{Var}_{x \sim \mathbb{N}(0,I)}(f(x)) &\leq \sum_{i=1}^{n} \mathbb{E}_{x \sim i} \mathrm{Var}_{x_i} f(x) \\
&\leq \sum_{i=1}^{n} \mathbb{E}_{x \sim i} \mathbb{E}_{x_i} (\partial_i f(x))^2 \\
&= \mathbb{E}_{x \sim \mathbb{N}(0,I)} \|\nabla f(x)\|_2^2
\end{aligned}$$

if we assume the thorem for $n = 1$.

Now, notice

$$\begin{aligned}
\mathrm{Var}_x f(x) &\approx_{CLT} \mathrm{Var}_z f\left(\frac{1}{\sqrt{k}}\sum_{i=1}^{k} z_i\right) \\
&\leq \frac{1}{2}\sum_{i=1}^{k} \mathbb{E}_{z,z'} f\left(\frac{1}{\sqrt{k}}\sum_{j\neq i} z_j + \frac{z_i}{\sqrt{k}}\right) \\
&= \frac{1}{4}\sum_{i=1}^{k} \mathbb{E}_z \left(f\left(\cdots + \frac{1}{\sqrt{k}}\right) - f\left(\cdots - \frac{1}{\sqrt{k}}\right)\right)^2 \\
&\approx_{taylor} \frac{1}{4}\sum_{i=1}^{k} \mathbb{E}_z \left(\frac{2}{\sqrt{k}} f'\left(\frac{1}{\sqrt{k}}\sum_{j\neq i}\right)\right) \\
&= \frac{1}{k}\sum_{i=1}^{k} \frac{1}{4}\mathbb{E}_z \left(\frac{2}{\sqrt{k}} f'\left(\frac{1}{\sqrt{k}}\sum_{j\neq i} z_j\right)\right) \\
&\approx_{CLT} \mathbb{E} f'(x)^2
\end{aligned}$$

Corollary 3.10
If $f : \mathbb{R}^n \to \mathbb{R}$ is L-Lipschitz, $|f(x) - f(y)| \leq L\|x - y\|_2$

$$\mathrm{Var}_x f(x) \leq L^2$$

Theorem 3.11
Let $W \in \mathbb{R}^{n\times n}$ be a symmetric random matrix with independent entries and assume $W_{ij} = W_{ji} \sim \mathbb{N}(\mu_{ij}, \sigma_{ij}^2)$. Then,

$$\mathrm{Var}(\lambda_1(W)) \leq 2\max_{i,j} \sigma_{ij}^2$$

3.2 Subgaussian Tail Bounds

Theorem 3.12 (Gaussian Concentration)
Let $x \sim \mathbb{N}(0, I)$, $F : \mathbb{R}^n \to \mathbb{R}$ be σ-Lipschitz (in 2-norm), then for all $t \geq 0$, $\mathbb{P}_x[|F(x) - \mathbb{E}[f]| \geq t] \leq 2e^{-\frac{t^2}{2\sigma^2}}$.

Proof 3.13
We will prove a slightly weaker bound of $2e^{-\frac{t^2}{\pi^2\sigma^2}}$. Assume F is $\mathcal{C}^1$ (otherwise we can "smoothen" it out). Then, for all x, $\|\nabla F(x)\|_2 \leq \sigma$. Without loss of generality, $\mathbb{E}[F] = 0$, otherwise, we will use $F - \mathbb{E}[F]$. We will prove the one-sided bound; the other side follows by using the function $-F$.
Now, we can use the exponential moment method to write

$$\Pr[F(x) \geq t] = \Pr[\exp(\lambda F(x)) \geq \exp(\lambda t)] \leq \frac{\mathbb{E}[\exp(\lambda F)]}{\exp(\lambda t)}$$

Let y be an independent copy of x.

$$\mathbb{E}[\exp(-\lambda F(y))] \geq \exp(-\mathbb{E}\lambda F(y)) = \exp(0) = 1$$

Then,

$$\mathbb{E}[\exp(\lambda(F(x) - F(y)))] = \mathbb{E}[\exp(\lambda F(x))]\mathbb{E}[\exp(-\lambda F(y))] \geq \mathbb{E}[\exp(\lambda F)]$$

The key trick is that

$$F(x) - F(y) = \int_0^{\pi/2} \frac{\partial}{\partial\theta} F(y\cos\theta + x\sin\theta)\,\mathrm{d}\theta$$

Calling $x_\theta = y\cos\theta + x\sin\theta$, note that $x_\theta \sim \mathbb{N}(0, I)$. Then, $x'_\theta = \frac{\partial x_\theta}{\partial\theta} = -y\sin\theta + x\cos\theta \sim \mathbb{N}(0, I)$. The two combinations are orthogonal hence x_θ and x'_θ are independent. Thus,

$$\begin{aligned}
\exp(\lambda(F(x) - F(y))) &= \exp\left(\lambda\frac{\pi}{2}\frac{1}{\pi/2}\int_0^{\pi/2}\frac{\partial}{\partial\theta}F(x_\theta)\,\mathrm{d}\theta\right) \\
&= \exp\left(\lambda\frac{\pi}{2}\mathbb{E}_{\theta\in[0,\pi/2]}[\frac{\partial}{\partial\theta}F(x_\theta)]\right) \\
&\leq \mathbb{E}_{\theta\in[0,\pi/2]}\exp\left(\lambda\frac{\pi}{2}\frac{\partial}{\partial\theta}F(x_\theta)\right) \\
&= \mathbb{E}_{x,y}\frac{2}{\pi}\int_0^{\pi/2}\exp\left(\lambda\frac{\pi}{2}\langle\nabla F(x_\theta), x'_\theta\rangle\right)\mathrm{d}\theta
\end{aligned}$$

Notice that $\langle\nabla F(x_0), x'_\theta\rangle$ is distributed as a Gaussian with variance at most σ^2, since the gradient is at most σ in magnitude and the two vectors are independent.

$$\begin{aligned}
\exp(\lambda(F(x) - F(y))) &\leq \frac{2}{\pi}\int_0^{\pi/2}\mathbb{E}_{x,y}\exp\left(\frac{1}{2}\left(\lambda\frac{\pi}{2}\sigma\right)^2\right)\mathrm{d}\theta \\
&\leq \exp\left(1/2(\lambda\pi/2\sigma)^2\right)
\end{aligned}$$

which implies our bound by choosing $\lambda = \left(\frac{2}{\pi}\right)\frac{t}{\sigma^2}$.

Corollary 3.14
Let $X = \sum_i g_i A_i$, $g_i \sim \mathbb{N}(0,1)$ independently, $A_i \in \mathbb{R}^{n\times n}$, symmetric, $\lambda_1(X)$ and $\|X\|_2 = \sigma_1(X)$ are both $\sigma_*(x)^2$-subgaussian, where

$$\sigma_*(x)^2 = \max_{\|v\|_2=1}\langle\sum_{i=1}^m A_i^{\otimes 2}, v^{\otimes 4}\rangle = \max_{\|v\|_2}\sum_{i=1}^m (v^T A_i v)^2$$

Proof 3.15
Let's say we want to prove this for λ_1. Then

$$\lambda_1(X(g)) - \lambda_1(X(h)) \leq \|X(g) - X(h)\|_2 = \left\| \sum_{i=1}^{m} (g_i - h_i) A_i \right\|_2 .$$

Then,

$$\sum_{i=1}^{m} (g_i - h_i) v^T A_i v \leq \left(\max_{\|v\|_2=1} \sum_{i=1}^{m} (v^T A_i v)^2 \right)^{1/2} \|g - h\|_2$$

and we can apply the previous theorem with this Lipschitz constant. (The spectral norm version follows from triangle inequality.)

Recall that previously, the error term was

$$\mathbb{E}_g \left\| \sum_i g_i A_i \right\|_2 \leq C\sqrt{\log n} \left\| \sum_i A_i^2 \right\|_2^{1/2}$$

Then, $\sigma(x)^2 = \max_{\|v\|_2=1} \sum_i v^T (A_i^2) v$. Now, our

$$\sigma_*(x)^2 = \sum_{i=1}^{m} \langle A_i v, v \rangle^2 \leq \sum_{i=1}^{m} \|A_1 v\|_2^2 \|v\|_2^2 = \sum_{i=1}^{m} v^T A_i^2 v = \sigma^2(x)$$

Corollary 3.16
Let $W \in \mathbb{R}^{n \times n}$, $W_{ij} = W_{ji} \sim \mathbb{N}(0, 1)$, $W_{ii} \sim \mathbb{N}(0, 2)$ (Wigner distribution). Then,

$$\Pr \|W\|_2 \geq \mathbb{E}\|W\| + t \leq 2e^{-t^2/4}$$

Theorem 3.17 (Talagrand Concentration)
Let $|X_i| \leq k$ be arbitrary bounded independent random variables, let $F : \mathbb{R} \to \mathbb{R}$, σ-Lipschitz, convex. then for all $t \geq 0$

$$\Pr_x [|F(x) - \mathbb{E}[f]| \geq tK] \leq 2e^{-t^2/2\sigma^2}$$

This will give tail bounds for $\|X\|_2$ with Rademachers, but not eigenvalues. Convexity is required, even over the hypercube.

Example 3.18
There is a 1-Lipschitz (but not convex), such that the above doesn't hold. On $\{0, 1\}^n$ take

$$f(x) = \sqrt{\max(-n/2 + \sum_i x_i, 0)}$$

Alternatively, Let $A = \left\{(x_1, \ldots, x_n) \in \{0, 1\}^n, \sum_i x_i \leq \frac{n}{2}\right\}$ and $f(x) = \inf_{y \in \{(x_1, \ldots, x_n) \in \{0,1\}^n, \sum_i x_i \leq \frac{n}{2}\}} \|x - y\|_2$. Obviously this Lipschitz, as this is distance function. Furthermore,

$$\Pr[|f - \mathbb{E}f| > n^{1/4}] = \Pr[|\sqrt{\max\{\frac{(2x_i - 1) + \cdots + (2x_n - 1)}{n}, 0\}} - \mathbb{E}[f]| > \sqrt{2}]$$

If the concentration held, the LHS would be $e^{-\sqrt{n}}$. But actually the thing inside the max looks like a Gaussian.

This means we are after

$$\Pr[|\sqrt{\max\{\zeta, 0\}} - \mathbb{E}f| > \sqrt{2}]$$

for some $\zeta \sim \mathbb{N}(0, I)$, but this is some constant.

4 Lecture 4

My laptop was dead.

5 Lecture 5

Given any hypergraph $H \subseteq \binom{[n]}{4}$ and $\{b_C\}$ is uniformly and indpendently random, we can certify/refute $\frac{1}{m}\sum_{C \in H} b_C x_C \leq \epsilon$, in $n^{O(\ell)}$ time so long as $m \geq \frac{n^2}{\ell}\frac{1}{\epsilon^2}\log(n)$. We did this by constructing the "Kikuchi matrix."

Suppose the Shannon Capacity satisfies $S(G) \geq 3n/4$. For any $\epsilon > 0$, there exists m such that $\alpha(G^m) \geq (3n/4 - \epsilon)^m$. Take the independent set X; consider removing

6 Lecture April 11

6.1 Spectral Sparsification

Recall last time we were trying to prove a sparsification result.

Theorem 6.1
Let $v_1, \ldots, v_m \in \mathbb{R}^n$ such that $\sum_i v_i v_i^T = I$. Then, $\forall \epsilon > 0$, there are $s_1, \ldots, s_m \in \mathbb{R}_+$, with at most $\lceil n/\epsilon^2 \rceil$ s_i's non-zero satisfying

$$(1-\epsilon)^2 I \preceq \sum_i s_i v_i v_i^T \preceq (1+\epsilon)^2 I$$

Note that this implies a graph sparsification result because we can take each v_i to be an edge Laplacian. The intuition is we start with $A = 0$ and add rank-1 updates to A, each of which is a $v_i v_i^T$. If we look at e.g. $x^T Ax = 1$, it starts as the empty set, then is the line, then a 2-d ellipse, etc. We hope to get a sphere at the end. We can actually track the evolution of polynomials very concretely. Let $\text{char}(A)$ be the characteristic polynomial of a matrix A.

$$\text{char}(A) = \det(xI - A) = \prod_i (x - \lambda_i)$$

Then for a rank-1 update:

$$\begin{aligned}
\text{char}(A + vv^T) &= \det\left(xI - A - vv^T\right) \\
&= \det(xI - A)\det\left(I - (xI - A)^{-1} vv^T\right) \\
&= \det(xI - A)(1 - \frac{\sum_i \langle u_i, v\rangle^2}{x - \lambda_i})
\end{aligned}$$

Then the eigenvalues interlace with the existing eigenvalues; i.e. notice that if any new eigenvalues are created, then this happens exactly when $\frac{\sum_i \langle u_i, v\rangle^2}{x-\lambda_i} = 1$, which happens between two λ_i.

Furthermore,

$$\mathbb{E}\left[\text{char}\left(A_0 + \sum_{i=1}^k v_i v_i^T\right)\right] = \left(1 - \frac{1}{m}\frac{\partial}{\partial x}\right)^k x^n$$

where the expectation is over the v_i. These are associated Laguerre polynomials, so we can confirm that the ratio of largest to smallest root is $(1+\epsilon)^2/(1-\epsilon)^2$ is $k \sim n/\epsilon^2$.

Today, we'll now control instead prove that using $6n$ terms,

$$I \preceq \sum_i s_i v_i v_i^T \preceq 13I.$$

At the beginning, all the eigenvalues are 0, so a lower barrier $\ell_0 = -n$ and an upper barrier $u_0 = n$ bound all eigenvalues. We will take $\sim n/\epsilon^2$ updates of $s_i v_i v_i^T$ in step i. If we take the lower barrier can be increased by $\delta_L = 1/3$ and the upper barrier can be increased by $\delta_U = 2$ to keep all the eigenvalues inside the barriers. Then this means at step $6n$, $-n + 6/3n = n \leq \lambda(A) \leq 13n = n + 12n$, so we get what we want by scaling. However, such a thing cannot work, because if the eigenvalues cluster near the barriers (i.e. large $\Omega(n)$-dimensional approximate eigenspaces), the eigenvalues cannot ALL move near the bottom (due to this interlacing observation), or might overshoot the top (a random vector has a large projection on a large approximate eigenspace). So we actually need to strengthen the inductive hypothesis to avoid clustering near the barriers. To do this, we introduce potential functions

$$\Phi^u(A) = \text{Tr}(UI - A)^{-1} = \sum_i \frac{1}{u - \lambda_i};\ \Phi_\ell(A) = \text{Tr}(A - \ell I)^{-1} = \sum_i \frac{1}{\lambda_i - \ell}$$

Notice that $\Phi^u(0) = 1$, so we will show that this stays smaller.

Lemma 6.2
We can always choose $+svv^T$ such that if $\Phi^u(A) \leq 1$, $\Phi_\ell(A) \leq 1$, then calling $u' = u + \delta_U$ and $\ell' = \ell + \delta_L$, then $\Phi^{u'}(A + svv^T) \leq 1$ and $\Phi_{\ell'}(A + svv^T) \leq 1$.

Proof 6.3
Recall the Sherman Morrison formula: $(A + vv^T)^{-1} = A^{-1} - \frac{A^{-1}vv^TA^{-1}}{1+v^TA^{-1}v}$. Then, $\mathrm{Tr}\big((A + vv^T)^{-1}\big) = \mathrm{Tr}\big(A^{-1}\big) - \frac{v^TA^{-2}v}{1+v^TA^{-1}v}$. Here, we will write A^{-1} even though we have pseudo-inverses (Alternatively, we could have just started with $A_0 = \epsilon I$). Then

$$\Phi^{u'}(A + svv^T) = \mathrm{Tr}\Big(u'I - A - svv^T\Big)^{-1}$$

$$\Phi^{u'}(A + svv^T) = \Phi^{u'}(A) + \frac{sv^T(u'I - A)^{-2}v}{1 - sv^T(u'I - A)^{-1}v}$$

$$\Phi^{u'}(A + svv^T) - \Phi^u(A) = (\Phi^{u'}(A) - \Phi^u(A)) + \frac{v^T(u'I - A)^{-2}v}{1/s - v^T(u'I - A)^{-1}v}$$

Rearranging, notice that the left-hand side is at most 0 if and only if for PSD matrix, $U_A = \Big((u'I - A)^{-1} + \frac{(u'I-A)^{-2}}{\Phi^u(A)-\Phi^{u'}(A)}\Big)$, we have

$$\frac{1}{s} \geq v^TU_Av$$

Similarly, there is a matrix L_A such that if $1/s \leq v^TL_Av$ and $\Phi_\ell(A)$ then $\Phi_{\ell'}(A + svv^T) \leq 1$. To see this, we compute

$$\Phi_{\ell'}(A + svv^T) = \Phi_{\ell'}(A) - \frac{v^T(A - \ell'I)^{-2}v}{1/s + v^T(A - \ell'I)^{-1}v}$$

$$\Phi_{\ell'}(A + svv^T) - \Phi_\ell(A) = (\Phi_{\ell'}(A) - \Phi_\ell(A)) - \frac{v^T(A - \ell'I)^{-2}v}{1/s + v^T(A - \ell'I)^{-1}v} \leq 0$$

Again, rearranging, $L_A = \frac{(A-\ell'I)^{-2}}{\Phi_{\ell'}(A)-\Phi_\ell(A)} - (A - \ell'I)^{-1}$ works.
The next claim is enough to prove the lemma, as

Lemma 6.4
There exists v such that $v^TU_Av \leq v^TL_Av$.

It's enough to prove that this is true for a v_i sampled at random, and they all add up to identity, so proving the inequality in expectation is the same as $\mathrm{Tr}(U_A) \leq \mathrm{Tr}(L_A)$.
Now, we have two subclaims:

- $\mathrm{Tr}(U_A) \leq \frac{1}{\delta_U} + 1$. To see this notice that

$$\mathrm{Tr}(U_A) = \frac{\mathrm{Tr}(u'I - A)^{-2}}{\Phi^u(A) - \Phi^{u'}(A)} + \Phi^{u'}(A)$$

  The first term is at most 1 by induction. Now, notice that the left term is exactly

$$\frac{\mathrm{Tr}(u'I - A)^{-2}}{\Phi^u(A) - \Phi^{u'}(A)} = \frac{-\frac{\partial}{\partial y}\Phi^y(A)\Big|_{y=u'}}{\Phi^u(A) - \Phi^{u'}(A)}$$

  Since $\Phi^y(A)$ is a convex function in y, then $\Phi^u(A) - \Phi^{u'}(A) \geq \delta_U(-\frac{\partial}{\partial u}\Phi^{u'}(A))$. This yields the inequality.

- $\mathrm{Tr}(L_A) \geq \frac{1}{\delta_L} - 1$. Now again,

$$\mathrm{Tr}(L_A) = \frac{\mathrm{Tr}\big((A - \ell' I)^{-2}\big)}{\Phi_{\ell'}(A) - \Phi_{\ell}(A)} - \Phi_{\ell'}(A)$$

  The second term is at least -1 by induction. Similarly, $\Phi_\ell(A)$ is a concave function, so we get the top as the derivative, so $\Phi^{\ell'}(A) - \Phi^{\ell}(A) \leq \delta_L(\frac{\partial}{\partial \ell}\Phi_{\ell'}(A))$

Finally, notice that for our choice of $\delta_L = 1/3$ and $\delta_U = 2$, we have $1/\delta_U + 1 = 3/2$ and $1/\delta_L - 1 = 2$, so we have our claim.

A work due to Zhu, Liao, and Orechhia use an algorithm called mirror descent.

6.2 Restricted Invertibility

Let L be a $n \times n$ real matrix L. It is invertible if and only if $x \neq 0 \implies Lx \neq 0$. For any matrix L, there exists a set $S \subseteq [n], |S| = \mathrm{rank}(L)$, then L_S is invertible. It turns out such a property is incredibly brittle.

For example, take a matrix of all 1's. If we add some noise to it, we can make its rank large; but it's wrong to say that this new matrix is strongly invertible.

Definition 6.5
$\sigma_{min}(L) = \min_{x \neq 0} \frac{\|Lx\|}{\|x\|}$.

Definition 6.6
L is well-invertible if $\sigma_{min}(L) \gg 0$.

Definition 6.7 (Stable Rank)

$$\mathrm{strank}(L) = \frac{\|L\|_F^2}{\|L\|_2^2} = \frac{\sum_i \sigma_i^2}{\sigma_{max}^2} \leq \mathrm{rank}(L)$$

Theorem 6.8 (Bourgain-Tzafriri 1987)
Given $n \times n$, L such that $\|Le_i\|_2 = 1$, for all i, then there exists $S \subseteq [n]$, $|S| \geq c\frac{n}{\|L\|_2^2}$ such that $\sigma_{min}(L_S) \geq d$ for some universal constants c, d.

We will prove essentially what is the tightest upgrade to this.

Theorem 6.9 (Vershynin '01, ..., Spielman-Srivastava '09)
Consider matrix L that's $n \times m$ with $m \geq n$ and $\epsilon > 0$. Then there exists S such that $|S| \geq c(\epsilon)\frac{\|L\|_F^2}{\|L\|_2^2}$ such that $\sigma_{min}^2(L_S) \geq d(\epsilon)\|L\|_F^2/m$.
In fact, $d(\epsilon) = 1 - \epsilon$ and $c(\epsilon) = \Theta(\epsilon^2)$.

Our proof outline is as follows. Define $L_{n\times m} = [W_1, \ldots, W_m]$. We build a matrix $A = L_S L_S^T$ one column at a time and a barrier b on σ_{min} which we will decrease in each step. At each step $A \leftarrow A + w_{i_1} w_{i_1}^T$. By interlacing, we can only introduce one new eigenvalue smaller than the existing ones; so we need to make sure our barrier b is under the

new one we introduce. Now we will run for k steps and $\lambda_k(A) \sim (1-\epsilon)^2 \frac{\|L\|_F^2}{m}$. Then, this will yield a $k \times k$ matrix with this well-invertibility property we'd like.

Suppose for the following that A has k nonzero eigenvalues, all $> b$.

Definition 6.10
w is b-good if $A + ww^T$ has $k+1$ eigenvalues has $> b$.

Lemma 6.11
If $w(A - bI)^{-1}w < -1$, then w is b-good.

Note that if add some w in the subspace spanned by the vectors we already added, then $(A - bI)^{-1}$ is PSD on this subspace, so this w cannot be b-good.

Proof 6.12
$\mathrm{Tr}(A - bI)^{-1} = \sum_{i \le k} \frac{1}{\lambda_i - b} - \frac{(n-k)}{b}$. Now, applying the rank-1 update,

$$\mathrm{Tr}\Big(A + ww^T - bI\Big)^{-1} = \sum_{i \le k} \frac{1}{\lambda_i' - b} + \frac{1}{\lambda_{k+1}' - b} - \frac{(n-k-1)}{b}$$

Subtracting these two terms yield:

$$\mathrm{Tr}\Big(A + ww^T - bI\Big)^{-1} - \mathrm{Tr}(A - bI)^{-1} = \sum_{i \le k} \left(\frac{1}{\lambda_i' - b} - \frac{1}{\lambda_i - b}\right) + \frac{1}{\lambda_{k+1}' - b} + \frac{1}{b}$$

Notice that the summation is at most 0 by interlacing. Thus, the LHS is at most $\frac{1}{\lambda_{k+1}' - b} + \frac{1}{b} = \frac{\lambda_{k+1}'}{b(\lambda_{k+1}' - b)}$. If we prove LHS is nonnegative, then we must also have $\lambda_{k+1}' - b$. To prove this, we write

$$\mathrm{Tr}\Big(A + ww^T - bI\Big)^{-1} = \mathrm{Tr}(A - bI)^{-1} - \frac{w^T(A - bI)^{-2}w}{1 + w^T(A - bI)^{-1}w}$$

Notice that the residual term is $-\frac{\text{PSD matrix}}{\text{negative number}}$, where the denominator follows by the assumption.

Observe that if $\sum_i w_i^T (A - bI)^{-1} w_i < -n$, then we're done (because a b-good vector always exists).

Definition 6.13
Define the potential as $\Phi_b(A) = \mathrm{Tr}\big(L^T(A - bI)^{-1}L\big)$.

Definition 6.14
w is a $b' = (b - \delta)$-great if it is b'-good and $\Phi_{b-\delta}(A + ww^T) \le \Phi(b)(A)$.

Lemma 6.15
If

$$w^T(A - (b-\delta)I)^{-1}LL^T(A - (b-\delta)I)^{-1}w \le (\Phi_b(A) - \Phi_{b'}(A))(-1 - w^T(A - b'I)^{-1}w)$$

then w is $(b - \delta)$-great.

Proof 6.16
Notice that the goodness is already ensured because $\Phi_b(A) \leq \Phi_{b'}(A)$ and the quantity on the left is the quadratic form of a PSD matrix, so $w^T(A - bI)^{-1}w < -1$. So, all we need to show is potential shift property.

$$\begin{aligned}
\Phi_{b'}(A + ww^T) &= \mathrm{Tr}\Big(L^T(A + ww^T - b'I)^{-1}L\Big) \\
&= \mathrm{Tr}\Big(L^T(A - b'I)^{-1}L\Big) - \frac{\mathrm{Tr}\big(L^T(A - b'I)^{-1}ww^T(A - b'I)^{-1}L\big)}{1 + w^T(A - b'I)^{-1}w} \\
&= \Phi_{b'}(A) - \frac{LHS}{1 + w^T(A - b'I)^{-1}w} \\
&= \Phi_b(A) - (\Phi_b(A) - \Phi_{b'}) - \frac{LHS}{1 + w^T(A - b'I)^{-1}w}
\end{aligned}$$

Rearranging, we find,

$$LHS = (\Phi_b(A) - \Phi_{b'}(A + ww^T) - (\Phi_b(A) - \Phi_{b'}(A)))(1 + w^T(A - b'I)^{-1}w)$$

By our assumption, this implies $(\Phi_b(A) - \Phi_{b'}(A + ww^T))(1 + w^T(A - b'I)^{-1}w) \leq 0$. This means that $\Phi_b(A) - \Phi_{b'}(A + ww^T) \geq 0$ as needed.

7 Lecture April 18

7.1 Beyond Matrix Khintchine [Bandeira-Boedchardjo-von Handel '21]

Recall the Matrix Khintchine inequality we started the class with.

Denote $X = \sum_i g_i A_i$ where $A_1, \ldots, A_n \in \mathbb{R}^{d \times d}$ symmetric and g_i are iid standard Gaussians. Then,

$$\mathbb{E}\left[\|X\|_2\right] \lesssim \sigma_0 \sqrt{\log d}$$

and $\sigma_0 = \left\|\sum_i A_i^2\right\|^{1/2}$. We also showed this is tight with some simple diagonal examples. But, in some cases we can remove the log factor (the expectation is always at least σ_0^2). If we force them to be very non-commutative, we can perhaps hope for a better bound.

Note that the distribution of the vector $\sum_i g_i A_i$ is jointly Gaussian. So, it has a well-defined covariance $d^2 \times d^2$ matrix. Let's give this a name.

Definition 7.1

$$\sigma_1^2 = \|\mathrm{Cov}\,(X)\|_2$$

Then,

Theorem 7.2 (BBvH'21)
Under the same hypotheses,

$$\mathbb{E}\left[\|X\|_2\right] \lesssim O(\sigma_0 + \log^{3/4}(d)\sigma_0^{1/2}\sigma_1^{1/2}) \leq O(\sigma_0 + \log^{3/2}(d)\sigma_1)$$

In fact, you can get the tight constant on σ_0 from the previous claim.

However, this bound is not always better. Let's try applying this new inequality to Kikuchi matrices. Take level ℓ Kikuchi matrices, $L_{C_1}, \ldots, L_{C_m}$ where C_i's are 4-uniform hyperedges. Here, the size of the matrices is $\binom{n}{\ell} \times \binom{n}{\ell}$ where ℓ is polynomial in n, so $m = poly(n)$ is exponentially smaller than the size of the matrices. However, for a given row S in $\sum_i a_i L_{C_i}$, it has correlation 1 with any T that it has an edge to. So, the covariance matrix will have really high spectral norm ($\binom{n-a}{\ell-b}$ neighbors for each vertex).

It's meaningful when for a given entry, there are very few entries we're correlated with (to make σ_1 small) and the size of the matrix is reasonable (smaller than or around the same as) number of the summands. The large matchings also correlate the entries together way too much.

For Wigner matrices, that is, a random symmetric matrix, then the histogram of the spectrum look like a semicircle, centered at 0 with radius $2\sqrt{n}$.

7.1.1 Discrepancy Minimization

We describe the simplest discrepancy minimization problem. Let $a_1, \ldots, a_n \in \mathbb{R}^d$ and all $\|a_i\|_\infty \leq 1$. The goal is to find $b_1, \ldots, b_n \in \{\pm 1\}$ such that $\|\sum_i b_i a_i\|_\infty \leq O(\sqrt{n}) \max\{1, \sqrt{\log \frac{d}{n}}\}$. Note: When you randomly choose the b_i, then $\|\sum_i b_i a_i\|_\infty \leq \sqrt{n}\sqrt{\log d}$ with probability $\Omega(1)$.

Theorem 7.3 (Spencer, 80s)
Goal is achievable. This is proved non-constructively via the entropy method.

We can try to generalize this to the matrix setting. $A_1, \ldots, A_n \in \mathbb{R}^{n\times n}$, $\|A_i\|_2 \leq 1$ for all i. The goal is to find $b_1, \ldots, b_n \in \{\pm 1\}$ such that $\|\sum_i b_i A_i\|_2 \leq O(\sqrt{n}\max\{\sqrt{\log d/n}, 1\})$. By matrix Khintchine, if we choose $b_1, \ldots, b_n$ uniformly at random, then with probability $\Omega(1)$, it's at most $O(\sqrt{n\log d})$. The conjecture is that this goal is acheivable, but we do not know how to prove it.

Today: we will show a paper by [Bansal-Jiang-Meka], the goal is achievable, if, in addition, $\|A_i\|_F^2 \leq \frac{n}{\log^3 n}$ for all i. For today, we will take $d = n$ and the maximum that the squared Frobenius norm becomes is n.

Concrete question: Suppose A_i's are adjacency matrices of perfect matchings. The operator norm is clearly 1, but the Frobenius norm is large. Is the Matrix Spencer conjecture true for this setting?

Lemma 7.4 (Partial Coloring, Rothvoss '14)
Let $\epsilon < 1/60000$, $\delta = \frac{3}{2}\epsilon\log_2\frac{1}{\epsilon}$. Let $H \subseteq \mathbb{R}^n$ be a subspace of dimension at least $\geq (1-\delta)n$. Let $K \subseteq H$ be a symmetric convex set such that $\gamma_H(K) \geq e^{-\delta n}$. Let $x_0 \in (-1,1)^n$. Then, there is a point $x \in (x_0+K) \subseteq [-1,1]^n$ and $|\{i \mid |x_i| = 1\}| \geq \frac{\epsilon n}{2}$.

Lemma 7.5
There are $c, c' > 0$ such that given $A_1, \ldots, A_n \in \mathbb{R}^{d\times d}$, symmetric such that $\left\|\sum_i A_i^2\right\|_{op} \leq \sigma^2$ and $\sum_i \|A_i\|_F^2 \leq nf^2$, given $x_0 \in (-1,1)^n$, there is an $x \in [-1,1]^n$ such that

$$\left\|\sum_{i=1}^n (x_i - x_{0,i})A_i\right\|_{op} \leq c(\sigma + \log^{3/4}(d)\sqrt{\sigma f})$$

and $|\{i \mid |x_i| = 1\}| \geq c'n$.

Consider the convex body $K = \{x \in \mathbb{R}^n : \|\sum_i x_i A_i\|_{op} \leq c_0\sigma + \log^{3/4}(d)\sigma^{1/2}f^{1/2}\} \subseteq \mathbb{R}^n$ for some constant c_0 we'll set later. To prove its measure is large, we can use BBvH concentration inequality. But first, we need to bound the covariance matrix

$$\mathrm{Cov}\,(X) = \sum_i \vec{A}_i\vec{A}_i^T \implies \mathrm{tr}(Cov(X)) \leq nf^2$$

But we have no idea what the largest eigenvalues are. But, we do know the number of eigenvalues of value at least $f^2/\delta =: \Delta^2$ is at most δn. This gives us a subspace $H = \{y \in \mathbb{R}^n : \sum_i y_i \mathrm{Tr}(A_iV_j) = 0, \forall j \in [k]\}$ where $V_1, \ldots, V_k \in \mathbb{R}^{d\times d}$ are the eigenvectors of $\mathrm{Cov}\,(()\,X)$ with eigenvalues at least Δ^2.

Now, let's prove the claim. Sample $g \sim \mathbb{R}^n$ by first sampling $y \in H$, $r \in H^\perp$, $g = y + r$. Now, let $Y = \sum_i y_i A_i$ and $R = \sum_i r_i A_i$, we have that $X = Y + R$ and $\mathbb{E}\left[X^2\right] = \mathbb{E}\left[Y^2\right] + \mathbb{E}\left[R^2\right]$. Then, $\sigma_0(Y) \leq \sigma_0(X) \leq \sigma_0$. Then, we know that for all $\vec{V}_j^T\,\mathrm{Cov}\,(Y)\,\vec{V}_j = 0$, then for all $v \perp \{V_1, \ldots, V_k\}$ then $v^T\,\mathrm{Cov}\,(Y)\,v \leq \Delta^2$. So,

$$\mathbb{E}\left[\|Y\|_{op}\right] \leq c(\sigma_0(Y) + \log^{3/4}(d)\sigma_0(Y)^{1/2}\sigma_1(Y)^{1/2}) \leq c'(\sigma_0 + \log^{3/4}(d)\sqrt{\sigma_0 f})$$

Define $K' = K \cap H$. Then, taking $c_0 = 2c'$, the probability we land in K' is thus at least $1/2$ by Markov, i.e. $\gamma_H(K') \geq 1/2$. Furthermore, by construction and our previous bound on the number of bad eigenvalues, the dimension of H is at least $(1-\delta)n$. Thus, by our lemma , there is $x \in (x_0 + K') \cap [-1,1]^n$ such that $|\{i \mid x_i \in \{\pm 1\}\}| \geq \epsilon/2$, proving the claim.

Recall in our original theorem, we had control over the Frobenius squared norm creating $f \leq \sqrt{\frac{n}{\log^3 n}}$, giving us the right answer by using the claim. Now, we can simply recurse the lemma over and over (by restricting the coordinates to those which are not set), getting a partial coloring each time. Furthermore, since we are setting $\Omega(n)$ coordinates each time, the error bound we get is $\sqrt{\text{number of coordinates left}}$, thus the total sum of operator residuals is something like $\sqrt{n} + \sqrt{cn} + \sqrt{c^2n} + \cdots = O(\sqrt{n})$.

7.1.2 A Proof of the Partial Coloring Lemma

Finally, let's prove Rothvoss' result.

Theorem 7.6
Let ϵ, δ be the same as in the Partial coloring lemma. Let $K \subseteq \mathbb{R}^n$ by a symmetric convex body, $\gamma_n(K) \geq e^{-\delta n}$. Let $x* \sim N(0,1)^n$ and $y^* \in K \cap [-1,1]^n$ minimizing $\|x* - y*\|_2$. Then, with probability $1 - e^{-\Omega(n)}$, $y*$ has $\geq \frac{\epsilon n}{2}$ coordinates set to ± 1. (Notice this gives an algorithm to find the point, as long as K has a polynomial time projection oracle).

Fact 7.7 (Sidak, Khatri '67, Special case of Gaussian Correlation Inequality)
If $K \subseteq \mathbb{R}^n$ symmetric, convex in $\mathbb{R}^n$ and let $S \subseteq \mathbb{R}^n := \{x \in \mathbb{R}^n \mid |\langle v, x\rangle| \leq \lambda\}$ be a strip. Then, $\gamma_n(K \cap S) \geq \gamma_n(K)\gamma_n(S)$.

Fact 7.8
Let $h(\epsilon) = \epsilon \log_2 \frac{1}{\epsilon} + (1-\epsilon)\log_2 \frac{1}{1-\epsilon}$. Then for all $0 < \epsilon < 1/2$, then the number of sets $|I| \leq \epsilon n$ is at most $2^{nh(\epsilon)} \leq 2^{3\epsilon/2 \log_2 1\epsilon}$.

Fact 7.9 (Gaussian Isoperimetric Inequality)
$K \subseteq \mathbb{R}^n$, H halfspace with $\gamma_n(K) = \gamma_n(H)$. Then, for all $\delta > 0$, $\gamma_n(K_\delta) \geq \gamma_n(H_\delta)$ where S_δ is the δ-net of K, i.e. the set of all points that are distance at most δ from S.

Corollary 7.10
Let $\epsilon > 0$. Then for all K, $\gamma_n(K) \geq e^{-\epsilon n}$, then $\gamma_n(K_{3\sqrt{\epsilon n}}) \geq 1 - e^{-\epsilon n}$.

To prove this corollary, just prove it for halfspaces! To get this volume, my halfspace must be distance $3/2\sqrt{\epsilon n}$ from the center, so if we expand to $3\sqrt{\epsilon n}$, we get most of the volume. Now we can prove the theorem. The intuition is that we can drop non-tight constraints from the solution y, and so the volume of the body will be much larger than the volume of $[-1,1]^n$.

Proof 7.11
Claim 1: $d(x*, [-1,1]^n) \geq c\sqrt{n}$ with probability $1 - e^{-\Omega(n)}$. Let I^* be the coordinate constraints which are tight (where $y^* = \pm 1$). Projecting to our set is the same as projecting to the set $K(I*) = K \cap \{x \in \mathbb{R}^n : |x_i| \leq 1, \forall i \in I*\}$. Suppose $|I*| < \epsilon n$. By Sidak/Khatri,

$$\gamma_n(K(I*)) \geq \gamma_n(K) \prod_{i \in I*} \gamma_n(\text{strip}) \geq e^{-\delta n} e^{-\epsilon n/2} \geq e^{-2\delta n}$$

Union-bounding over all $I*$ we get $\gamma_n(\cup_{I*} K(I*)) \leq e^{-\delta n}$. Thus, $\gamma_n(K(I*)_{3\sqrt{2\delta n}}) \geq 1 - e^{-\delta n}$ and we can set δ small enough.

8 Lecture April 25

8.1 Concentration of Wigner Matrices

Definition 8.1 (Wigner Matrices)
The Wigner distribution is the distribution of symmetric matrices $A \in \mathbb{R}^{d\times d}$ where entries on and above the diagonal are independent $N(0,1)$. Formally, defining $A_{ij} = e_i e_j^T + e_j e_i^T$, $i \neq j$, $A_{ii} = e_i e_i^T$ and $g_{ij} \sim N(0,1)$ iid,

$$A = \sum_{1\leq i\leq j\leq d} g_{ij} A_{ij}$$

By Matrix Khintchine, we have found that $\mathbb{E}\left[\|A\|_2\right] \lesssim \sqrt{d\log d}$. However, we will prove the following theorem.

Theorem 8.2
For A distributed with the Wigner distribution on d by d matrices,

$$\mathbb{E}\left[\|A\|_2\right] \leq 2\sqrt{d} + o(\sqrt{d})$$

Proof 8.3
Call the eigenvalues of A as $\lambda_1 \geq \cdots \geq \lambda_d$. Take k even. Then,

$$\lambda_{max}^k \leq \operatorname{tr}\left(A^k\right) = \sum_i \lambda_i^k \leq d\lambda_{max}^k$$

Furthermore,

$$\lambda_{max} \leq \operatorname{tr}\left(A^k\right)^{1/k} \leq d^{1/k}\lambda_{max}$$

This lets us approximate the top eigenvalue with the k-th trace moment, which is the quantity we sandwich in the above inequality.

$$\mathbb{E}\left[\operatorname{tr}(A)\right] = 0, \mathbb{E}\left[\operatorname{tr}\left(A^2\right)\right] \sim d^2$$

So already, we get that $\lambda_{max} \in [\sqrt{d}, d]$. To compute higher moments, we can use combinatorics

$$\mathbb{E}\left[\operatorname{tr}\left(A^4\right)\right] = \sum_{i_1,i_2,i_3,i_4} \mathbb{E}\left[A(i_1,i_2)A(i_2,i_3)A(i_3,i_4)A(i_4,i_1)\right]$$

Thinking of A as an undirected weighted graph, notice that thinking about (i_1,i_2,i_3,i_4,i_1) as a circuit in the complete graph on d vertices, if any "edge" appears an odd number of times, the expectation of this term is 0. For the case of $k = 4$, we could have:

1. three distinct vertices, and two distinct edges. Then, we have $\Theta(d^3)$ ways to choose the three distinct vertices, and $\Theta(d^3)$ ways to choose up to cyclic permutation as well. Each term is contributing $\mathbb{E}\left[g_{ij}^2\right]\mathbb{E}\left[g_{ij}^2\right] = 1$ so this adds $\Theta(d^3)$ to the sum.

2. two distinct vertices and the same edge appearing four times. This is only $\Theta(d^2)$.

Thus, $\mathbb{E}\left[\operatorname{tr}\left(A^4\right)\right]^{1/4} \sim d^{3/4}$, so we get $\lambda_{max} \in [d^{1/2}, d^{3/4}]$.
With a handle on how to compute such moments, let's try to find

$$\mathbb{E}\left[\operatorname{tr}\left(A^k\right)\right] = \mathbb{E}\sum_{i_1,\ldots,i_k} A(i_1,i_2)\ldots A(i_k,i_1)$$

We define a "non-crossing pair" partition of the edges as a pairing of edges so that the walk $(i_1, i_2, \ldots, i_k, i_1)$ has $k/2$ distinct edges and $k/2 + 1$ vertices. This is called "non-crossing" because if we were to represent the edges on points $p_{12}, p_{23}, \ldots$ one-dimensional number line, the pairing is a way of drawing arcs below the number line between pairs of $p_{i,i+1}$ such that none of the arcs cross each other. The way to see this relationship is the view that the first kind of walk is exactly a tree; since we repeat everything twice, this is exactly a DFS walk. Furthermore, the pre and post-order numbers can be the embedding on the number line, where we pair pre and post-order numbers of the root of every edge. These will be the leading terms in our expansion.
The number of distinct vertex choices is $d^{k/2} + 1$. Furthermore, if we want to choose which edges are paired up, we have $\leq k^{k/2}$ ways, just k choices for edge 1's pair, $k - 1$ choices for edge 2's pair.

Lemma 8.4
For all k, $\mathbb{E}\left[\operatorname{tr}\left(A^k\right)\right] \leq d^{k/2+1} C_k$ where C_k is the kth Catalan number.

This can be done by thinking about the number of DFS walks at Catalan numbers. Furthermore, $C_k \leq 2^k$, so we get $(2 + o(1))\sqrt{d}$ as desired.

In fact, the tighter estimate implies the so-called semicircle law. With a matrix A, we can associate the distribution

$$\frac{1}{\sqrt{d}} A \to \frac{1}{d} \sum_{i=1}^{d} \delta_{\lambda_i}$$

where δ_{λ_i} is just the indicator on λ_i. In fact, as $d \to \infty$, we get that the distribution goes to a semicircle, i.e. $\sqrt{1 - x^2}$ for $x \in [-2, 2]$. To see this, you will get estimates on $\mathbb{E}\left[\operatorname{tr}\left(A^k\right)\right]^{1/k} \sim C_k$. But, the moments of the semicircle are also the catalan numbers!

8.2 Beyond Matrix Khintchine [BBvH'23]

Theorem 8.5
Suppose A_i are $d \times d$, self-adjoint, real matrices and $X = \sum_i g_i A_i$, $\sigma(X) = \left\|\sum_i A_i^2\right\|^{1/2}$. By Matrix Khintchine, we know that $\mathbb{E}\left[\|X\|\right] \lesssim \sqrt{\log d}\sigma$. Further, defining $\nu(X) = \|\operatorname{Cov}(X)\|^{1/2}$,

$$\mathbb{E}\left[\|X\|\right] \lesssim \sigma(X) + c\nu(X)^{1/2}\sigma(X)^{1/2}\log^{3/4}(d) \lesssim \sigma(x) + c'\nu(x)\log^{3/2}(d)$$

There is a better bound that's always better than Khintchine, by using a so-called alignment parameter $W(X)$. However, it wasn't very useful as this parameter is hard to compute. The innovation of BBvH was introducing the ν parameter as an upper bound for W, which is much easier to compute.

$$\mathbb{E}\left[\operatorname{tr}\left(A^4\right)\right] = \sum_{i_1, i_2, i_3, i_4} \operatorname{tr}\left(A_{i_1} A_{i_2} A_{i_3} A_{i_4}\right)$$

The only contributing terms are ones that look like $\operatorname{tr}(A_1 A_2 A_1 A_2)$ and $\operatorname{tr}\left(A_1^2 A_2^2\right)$. However, if we expect non-commutative, the difference between crossing the partitions (the former) and non-crossing partitions (the latter) should be large. This motivates defining

$$W(X) = \sup_{U, V, W \text{ orthogonal}} \sum_{i,j} \operatorname{tr}\left(A_i U A_j V A_i W A_j\right)$$

(the intuitive notion is the one without the orthogonal transformations.) Tropp proved $\mathbb{E}\left[\|X\|\right] \lesssim \log^{1/4}(d)\sigma(X) + \log^{1/2}(d)W(X)$. Furthermore, $W \leq \sigma$.

Here is a proof sketch of the later result. We will relate $\sum_i g_i A_i$ to $\sum_i G_i^N \otimes A_i$, where G_i^N is a $N \times N$ symmetric matrix with all entries on and above the diagonal have variance $1/N$. For intuition we can consider $\mathbb{E}\left[\operatorname{tr}\left(G_i G_j G_i G_j\right)\right]$

and $\mathbb{E}\left[\operatorname{tr}\left(G_i^2 G_j^2\right)\right]$ (as any other trace moment will vanish). It turns out that $\frac{1}{N}\mathbb{E}\left[\operatorname{tr}\left(G_i G_j G_i G_j\right)\right] \to 0$ as $N \to \infty$. However, $\frac{1}{N}\mathbb{E}\left[\operatorname{tr}\left(G_i^2 G_j^2\right)\right] \to c$. Now, we took a tensor product so that the "non-commutativity" of G contributes largely to the model.

$$\mathbb{E}\left[\left\|\sum_i G_i^N \otimes A_i\right\|\right] \leq 2\sigma + o_N(1)$$

Now, to relate the two quantities, look at $\sum_i D_i^N \otimes A_i$ where D_i^N is a N by N diagonal matrix of independent standard Gaussians. Notice that $\left\|D_i^N\right\|_F = \left\|G_i^N\right\|_F$. The normalized trace of this model will be the same as in the original model, as the blocks don't influence each other and we're averaging. We will interpolate between D_i^N and G_i^N and bound the derviative of the trace moments during this process.

"Free Probability" is where we can make abstract variables $S_1, \ldots, S_k$ for each matrix. Another way to think about this is there is an algebra of these random variables, then a probability measure assigns expectations monomials. The trick to make this work with matrices is making a non-commutative polynomial of these abstract variable, and giving a trace functional where we assign for example, saying a crossing pair partition can be set to 0. So we can also use this view to compute $\mathbb{E}\left[\left\|\sum_i G_i^N \otimes A_i\right\|\right]$.

Then, the interpolation scheme is

$$X_q^N = \sum_{i=1}^{n} (\sqrt{q} D_i^N + \sqrt{1-q} G_i^N) \otimes A_i$$

Now, we want to bound

$$\frac{\mathrm{d}}{\mathrm{d}q}\mathbb{E}\left[\operatorname{tr}\left((X_q^N)^k\right)\right]$$

It turns out there is a formula for a function of this average thing.